

Java Methods & Constructors Practice

Basic Java Practice Questions

1. METHODS

1. Program to calculate the area of a circle.

Scenario: Create a Java program with a method that accepts the radius of a circle and calculates its area.

Input:

```
Enter radius: 7
```

Output:

```
Area of Circle: 153.86
```

2. Program to calculate the total and average marks.

Scenario: Create a Java program with a method that accepts marks of three subjects and calculates the total and average marks.

Input:

```
Enter marks of Subject 1: 80  
Enter marks of Subject 2: 75  
Enter marks of Subject 3: 90
```

Output:

```
Total Marks: 245  
Average Marks: 81.66
```

3. Program to convert Celsius to Fahrenheit.

Scenario: Create a Java program with a method that accepts temperature in Celsius and converts it into Fahrenheit.

Formula:

```
Fahrenheit = (Celsius × 9/5) + 32
```

Input:

```
Enter temperature in Celsius: 25
```

Output:

```
Temperature in Fahrenheit: 77.0
```

2. CONSTRUCTORS

1. Create a Book class using a parameterized constructor.

Scenario: Create a Book class with book title, author name, and price. Use a parameterized constructor to initialize the book details and display them.

Input:

```
Enter book title: Java Programming
Enter author name: James
Enter price: 450
```

Output:

```
Book Details
Title: Java Programming
Author: James
Price: 450
```

2. Create a Mobile class using a parameterized constructor.

Scenario: Create a Mobile class with brand name, model name, and storage capacity. Use a parameterized constructor to initialize the details and display them.

Input:

```
Enter brand: Samsung
Enter model: S24
Enter storage: 256
```

Output:

```
Mobile Details
Brand: Samsung
Model: S24
Storage: 256 GB
```

3. Create a Product class using constructors to initialize product details.

Scenario: Create a Product class with product name, product ID, and price. Use a parameterized constructor to initialize the product details and display them.

Input:

```
Enter product name: Laptop
Enter product ID: 501
Enter price: 55000
```

Output:

```
Product Details
Product Name: Laptop
Product ID: 501
Price: 55000
```